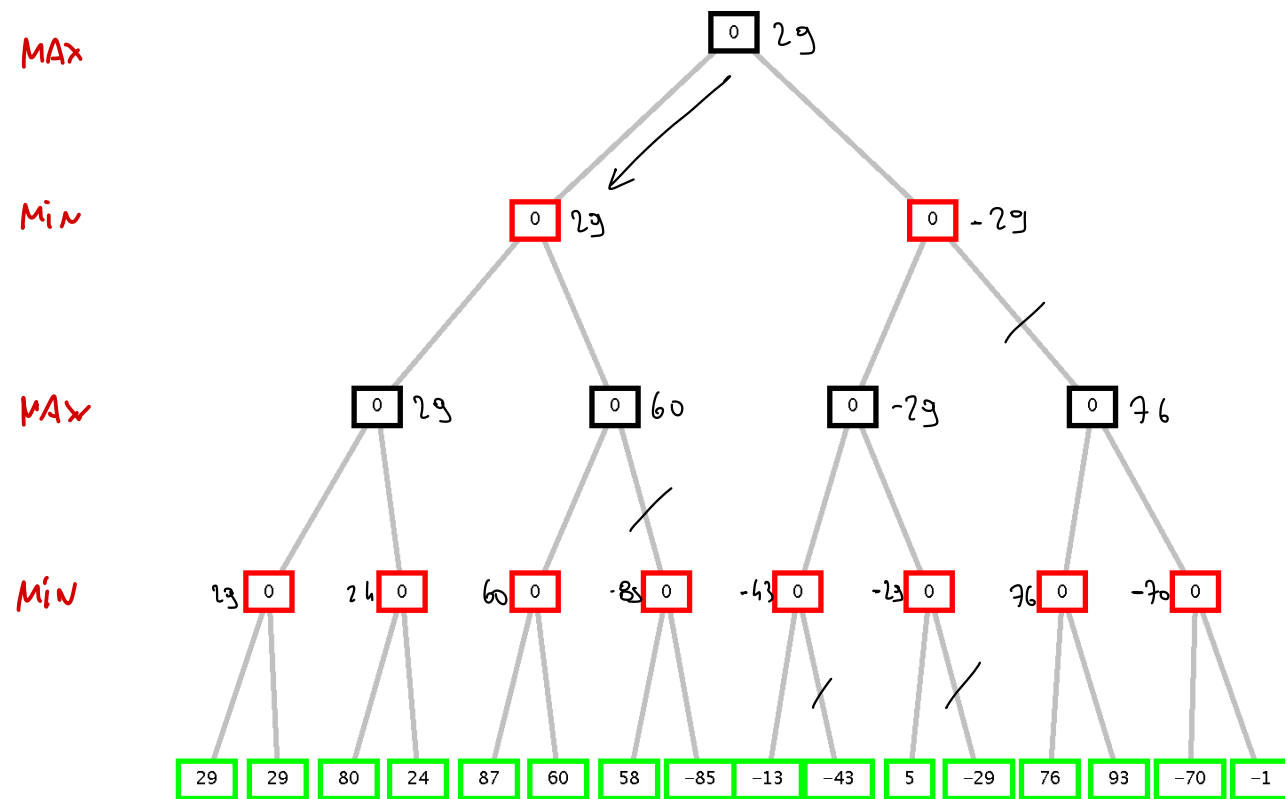


EXAMPLE EXAM OF FUNDAMENTALS OF AI - FIRST MODULE

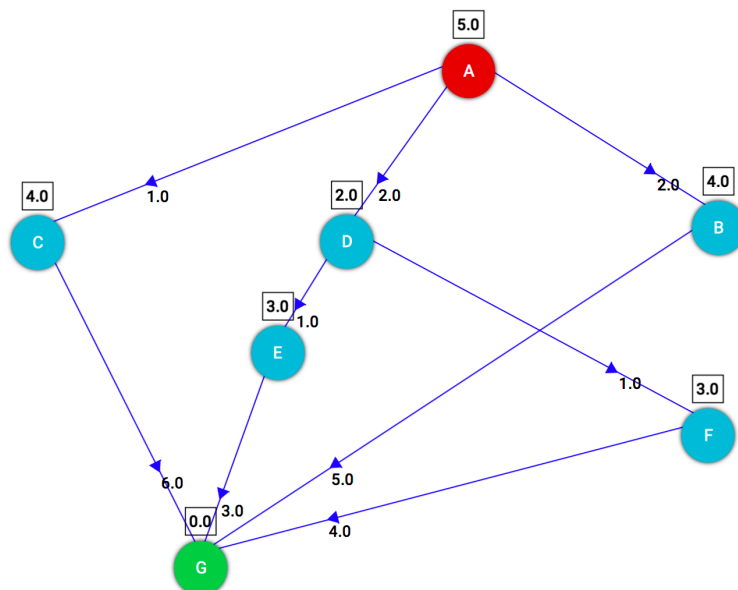
Exercise 1

Consider the following game tree where the first player is *MAX*. Show how the *min-max* algorithm works and show the *alfa-beta* cuts. Also, show which is the proposed move for the first player.



Exercise 2

Consider the following graph, where A is the starting node and G the goal node. The number on each arc is the cost of the operator for the move. Close to each node there is the heuristic evaluation of the node itself, namely its estimated distance from the goal:



- Show the search tree generated by the A* algorithm along with the order of expansion of nodes. In case of ties, chose the node to expand in alphabetical order.
- Is the heuristic admissible?
- Which is the cost of the path found by A* and the number of nodes expanded?

Exercise 3

Given the following CSP:

A::[1, 2, 3, 4, 5, 6]

B::[1, 2, 3, 4, 5, 6]

C::[1, 2, 3, 4, 5, 6]

$A \geq B + 1$

$B \geq C - 3$

Find the first solution through tree search, by applying forward checking, using alphabetical order of variables and lexicographic order of values.

Exercise 4

Given the following initial state **[at(locationA), have_battery, handempty]**: and actions modeled as follows:

take_picture(Location)

PRECOND: have_battery, at(Location), have_camera

DELETE: have_battery

ADD: picture(Location)

putdown_camera

PRECOND: have_camera

DELETE: have_camera

ADD: handempty

pickup_camera

PRECOND: handempty

DELETE: handempty

ADD: have_camera

charge_battery

PRECOND: not have_battery

DELETE: -

ADD have_battery

go(Location1, Location2)

PRECOND: at(Location1)

DELETE: at(Location1)

ADD at(Location2)

and the following goal **picture(locationA), picture(locationB)**

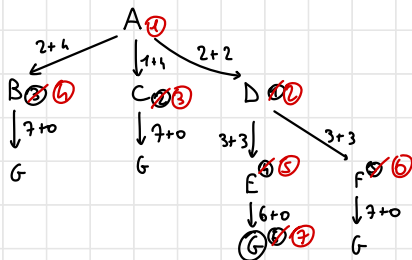
Solve the problem by using the POP algorithm showing threats and how to solve them.

Exercise 5

- 1) Model the action **take picture** (preconditions, effects and frame axioms), the initial state and the goal of the exercise 4 using the Kowalsky formulation
- 2) Show two levels of graph plan when applied to exercise 4.
- 3) What is conditional planning and what are its main limitations?
- 4) What is ant colony optimization?
- 5) What are the main features of iterative deepening?

Es. 2

a)



c) tot cost = 6

b) yes, all paths from A to G has cost ≥ 5

Es. 3

$$A = 1 \xrightarrow{A \geq B+1} \times$$

$$A = 2 \xrightarrow{A \geq B+1} B :: [1]$$

$$B = 1 \xrightarrow{B \geq C-3} C :: [1, 4]$$

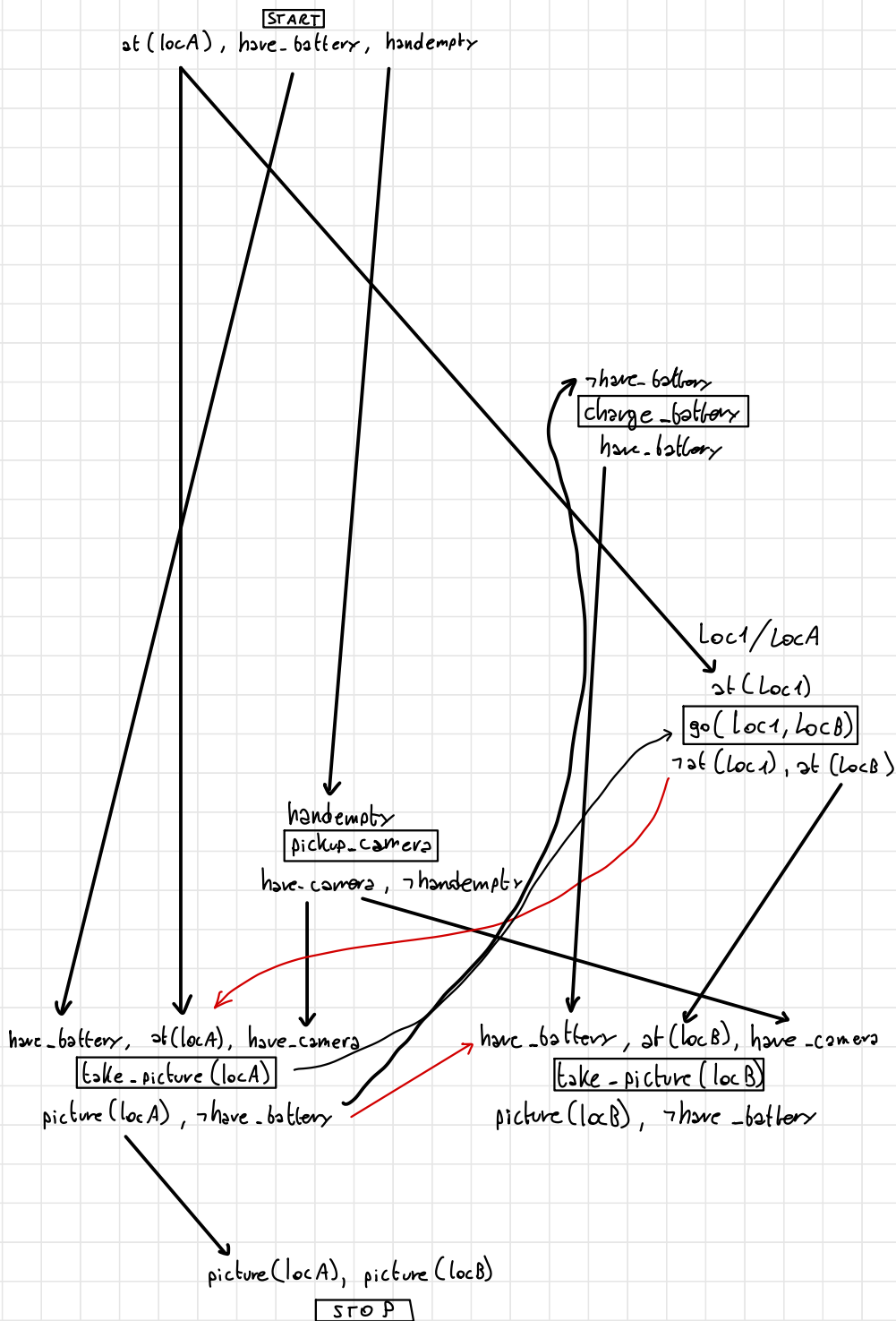
$$C = 1$$

solution: $A = 2$

$$B = 1$$

$$C = 1$$

Es. 4



$\langle \text{START}, \text{take_picture}(\text{locA}), \text{at}(\text{locA}) \rangle$ is threatened by $\text{go}(\text{locA}, \text{locB})$, we solve it by promoting $\text{go}(\text{locA}, \text{locB})$

$\langle \text{START}, \text{take_picture}(\text{locB}), \text{have_battery} \rangle$ is threatened by $\text{take_picture}(\text{locA})$, we solve it by introducing white knight charge_battery to restore have_both .

ES. (5.1)

INITIAL STATE:

$\text{holds}(\text{at}(\text{locA}), s \emptyset)$.

$\text{holds}(\text{have_battery}, s \emptyset)$.

$\text{holds}(\text{handempty}, s \emptyset)$.

EFFECTS

$\text{holds}(\text{picture}(\text{location}), \text{do}(\text{take_picture}(\text{location}), s))$.

PRECONDITIONS:

$\text{pact}(\text{take_picture}(\text{location}), s) :-$

$\text{holds}(\text{have_battery}, s),$

$\text{holds}(\text{at}(\text{location}), s),$

$\text{holds}(\text{have_camera}, s).$

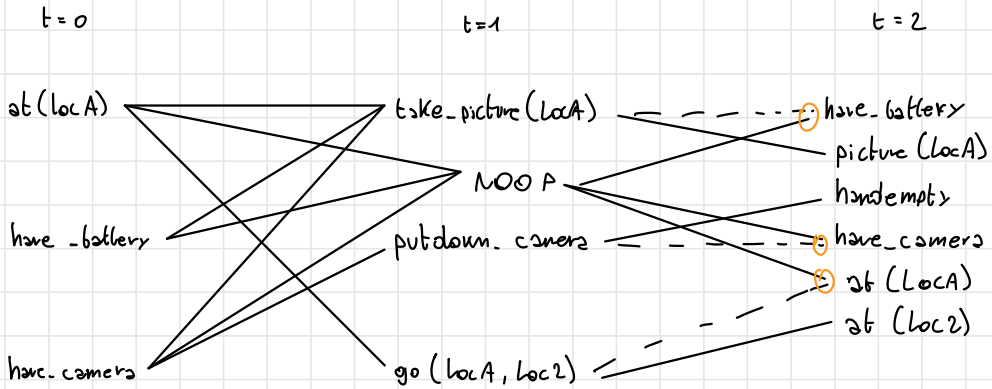
FRAME AXIOMS:

$\text{holds}(V, \text{do}(\text{take_picture}(\text{location}), s)) :-$

$\text{holds}(V, s),$

$V \neq \text{have_battery}.$

ES. (5.2)



noop on have_battery and take_picture(locA)

noop on have_camera and putdown_camera

noop on at(locA) and go(locA, loc2)

at(locA) and at(loc2)

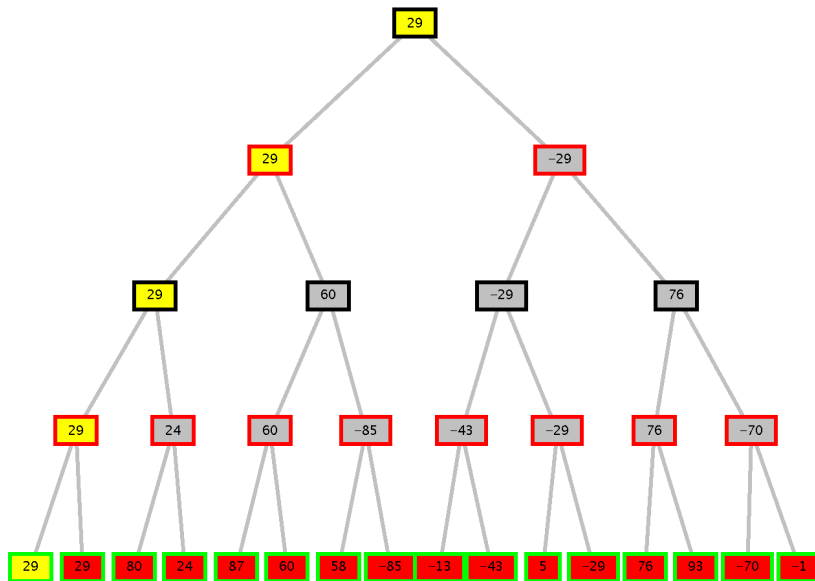
have_battery and picture(locA)

handempty and have_camera

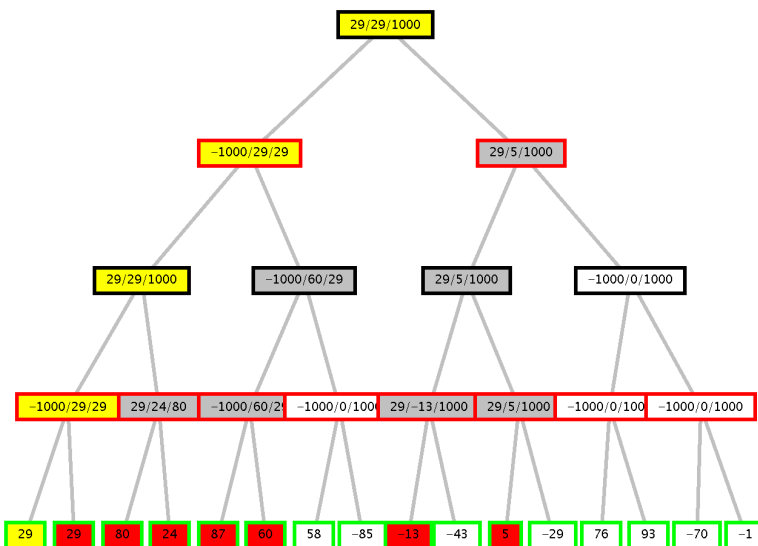
Solution

Exercise 1

Min-max

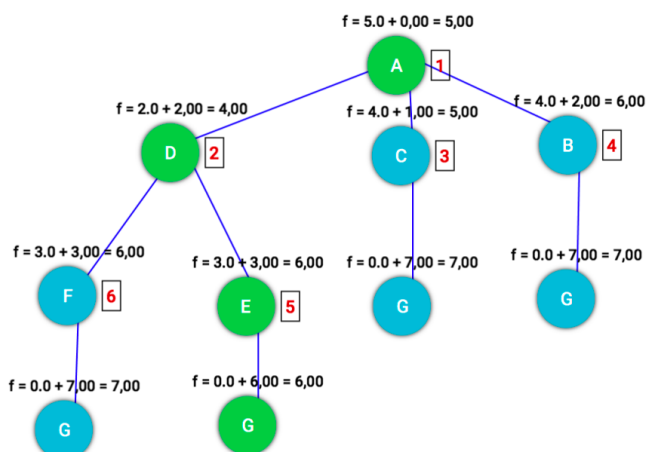


Alfa-beta cuts:



Exercise 2

A* (Admissible heuristics; cost of the path: 6; number of expanded node: 6, with a square box close to them with the order of expansion)

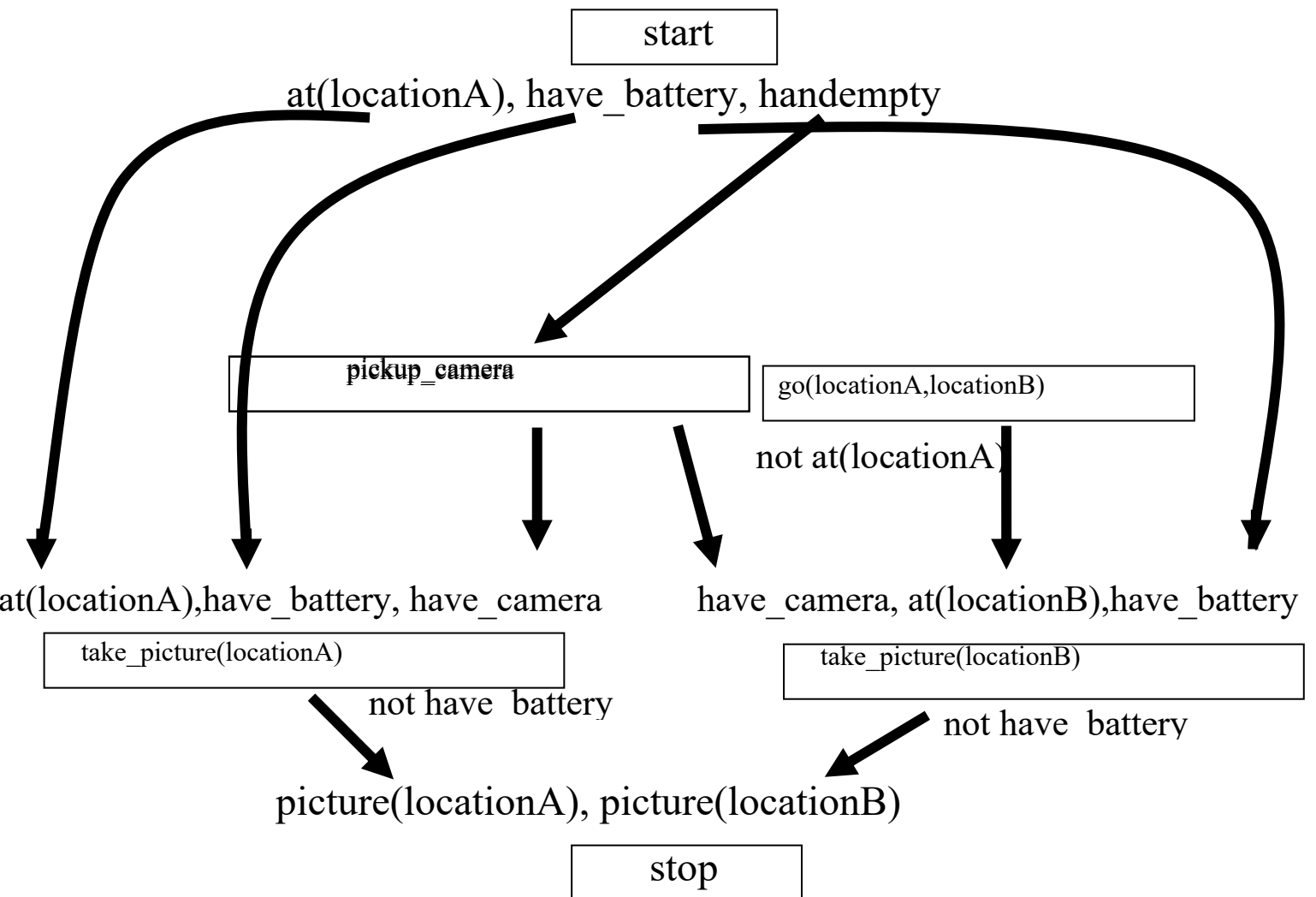


Exercise 3

Backtracking
Labeling e FC
Labeling e FC
Labeling e FC

A	B	C
A=1	Fail	[1...6]
A=2	[1]	[1...6]
A=2	B=1	[1...4]
A=2	B=1	C=1

Exercise 4



The plan up to now contains threats. In particular:

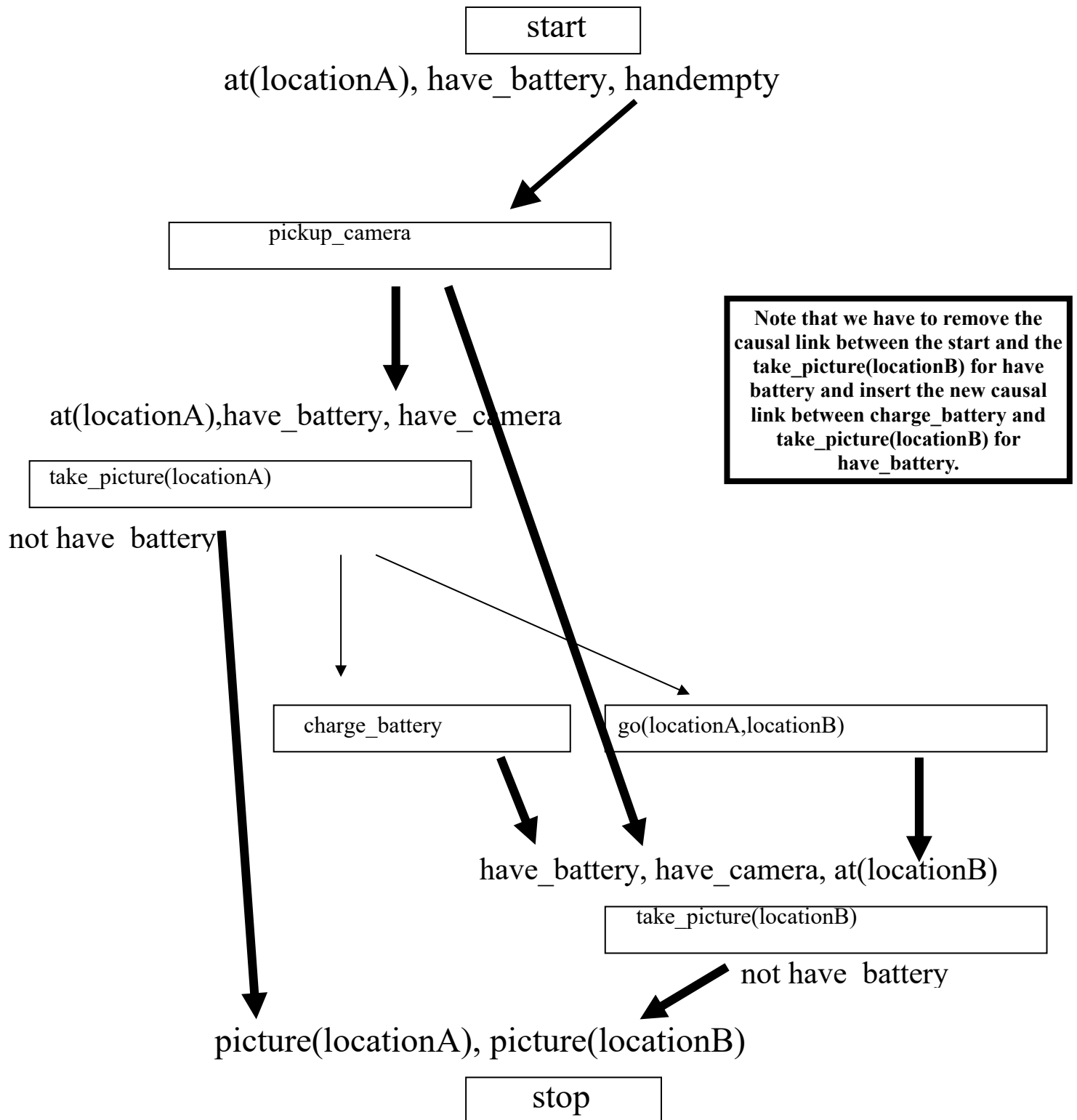
<Start, take_picture(locationB), have_battery> and <Start, take_picture(locationA), have_battery> are threatened by take_picture(locationA) and take_picture(locationB) respectively.

No ordering constraints can solve these threats: we need to insert a white knight charge_battery.

In addition the action go(locationA, locationB) threatens causal link

<Start, take_picture(locationA), at(locationA)>

In this case demotion can solve the threat. We introduce an ordering constraint between take_picture(locationA) and go(locationA, locationB).



Exercise 5

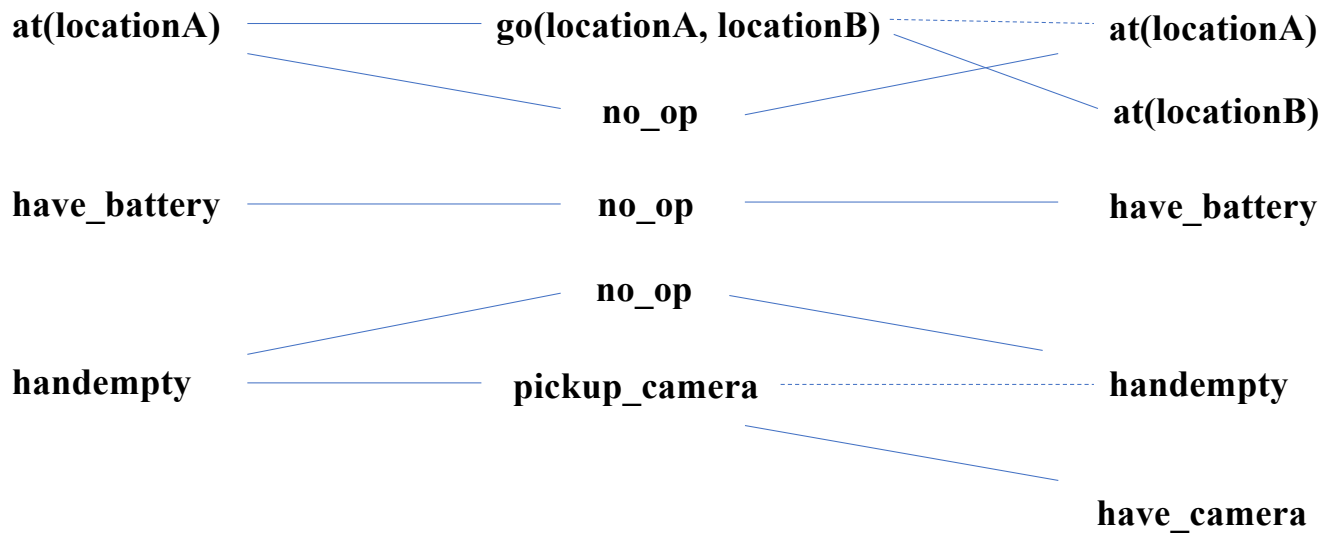
1)

```

holds(at(location1),s0).
holds(have_battery, s0).
holds(handempty,s0).
holds(picture(Location), do(take_picture(Location),S))
pact(take_picture(Location),S):- holds(have_battery, S), holds(at(Location),S), holds(have_camera,S).
holds(V,do(take_picture(Location),S)):- holds(V,S), V\=have_battery.

```

2)



go(locationA,locationB) incmpatible with no_op on at(locationA)

handempty and have_camera are incompatible

at(locationA) and at(locationB) are incompatible